IS1111 Tutorial Assignment 6 – Dictionaries, Files & CRUD

Overview

This lab gives you practice with dictionaries, converting between lists and dictionaries, and working with files (TXT, CSV, JSON). You will also build CRUD features: Create, Read, Update, Delete.

- Dicts: building dictionaries, safe lookup, and dict/list conversions
- Files: writing/reading TXT, CSV, and JSON:
  - .txt file stores plain text with no built-in structure, so it's just lines of characters
  - .csv file – stores data in rows and columns separated by commas
  - .json file stores structured data using key–value pairs (like dictionaries and lists in Python)
- CRUD patterns: Create, Read, Update, Delete
- Simple menu loops and saving changes back to a file

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1. Dict and List conversions

You are given a list of module codes:

```
modules = ["IS1111", "IS1109", "IS1103", "IS1120"]
```

(a) Create an empty dictionary called module_lengths.

(b) Loop through the list and store:

- Key = module code
- Value = length of module code

(c) Convert the dictionary into a list of tuples using .items() and print it.

(d) Sort the tuples by length (highest first) and print the sorted result.

HINT: You may need to create tuples like (length, code).

2. TXT file + CRUD basics

Use a text file called "shopping.txt".

(a) **CREATE**: ask the user for 3 shopping items and store them in a list.

HINT: Use the following loop and include underscore "_" : for _ in range(3):

(b) **SAVE**: write the list to shopping.txt (one item per line).

HINT: Use open("shopping.txt", "w")

(c) **READ**: open shopping.txt and print all items neatly.

(d) **UPDATE**: ask the user for one extra item and append it to the file.

(e) **DELETE**: ask the user for an item name to remove. Read all items, remove it if found, then write the updated list back.

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3. CSV file + dictionary records

You are given a dictionary of products:

```
products = {
    "P001": {"name": "Laptop", "price": 900},
    "P002": {"name": "Mouse", "price": 25},
    "P003": {"name": "Keyboard", "price": 45},
}
```

Use a CSV file called "products.csv" with this header row:

```
id,name,price
```

(a) **CREATE**: write the products dictionary into products.csv.

HINT: First write a header row: id,name,price

HINT: Loop through dictionary using: for pid, data in products.items():

(b) **READ**: read products.csv and rebuild a new dictionary called products_from_file (same structure).

(c) **UPDATE**: ask for a product ID and a new price. Update if found, otherwise print "Product not found".

(d) **SAVE**: write the updated products_from_file back to products.csv.

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4. JSON + full CRUD menu system

Use a JSON file called "students.json".

In Python, your data should be a LIST of DICTIONARIES like this:

```
students = [
    {"id": 1, "name": "Sarah", "grade": 75},
    {"id": 2, "name": "Ben", "grade": 60},
    {"id": 3, "name": "Aoife", "grade": 82},
]
```

(a) **CREATE FILE**: save the starter list into students.json.

(b) Build a menu loop with these options:

```
1. View students
2. Add student
3. Update grade
4. Delete student
5. Exit
```

(c) **READ**: View students loads students.json and prints all students neatly.

(d) **CREATE**: Add student asks for name + grade, generates the next id, saves back to JSON.

(e) **UPDATE**: Update grade asks for student id, updates if found, saves back to JSON.

(f) **DELETE**: Delete student asks for student id, removes if found, saves back to JSON.

(g) Calculate average grade.

HINT: Use a loop to total grades, then divide by len(students).